XFS4IoT SP-Dev Workgroup

3rd March 2026

© 2026 KAL ATM Software GmbH (KAL)

Confidential

XFS4IoT SP-Dev Workgroup agenda

- Recap previous meeting
- Framework update
- DK protocol update
- Guest speaker - Synaforce
- What’s next?
- Next meeting

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL’s written permission

Recap meeting December 2025

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL's written permission

Recap previous meeting

- Support for .NET 10 (Released November 2025)
  - Why it is important to stay on the latest version
  - Impact on SP-Dev Framework
- SP-Dev Framework on ‘embedded’ hardware
  - Hardware used
  - KAL experiments
  - Findings and next steps

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL’s written permission

Framework update

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL's written permission

Summary

1. New framework version 3.3.0 is available

1.1. Added German interface in the framework

1.2. Removed reflection for the command handler to reduce memory usage

1.3. Sample PinPad updated - German interface example

https://github.com/KAL-ATM-Software/KAL_XFS4IoT_SP-Dev/releases/tag/v3.3.0

2. New SP template package version 3.3.0 is available

2.1. Added new PinPad SP template project using German interface

https://github.com/KAL-ATM-Software/KAL_XFS4IoT_SP-Dev-Samples/releases/tag/v3.3.0

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL's written permission

DK protocol (German Interface)

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL's written permission

German DK interface

- DK = Deutsche Kreditwirtschaft = GBIC = German Banking Industry Committee

- DK topic was covered in the 36th Workshop, 3rd September 2024
- Thomas from SBS presented German DK information
- DK is known as ZKA (Zentraler Kreditausschuss) until 2011
- GBIC publishes a set of rules and procedures for the German ATM system
- DK is to ensure that the banking client has the same high standard of security and service on all participating ATMs
- Homepage: https://die-dk.de

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL's written permission

German DK interface main features

- **OPT**

Online Personalisierung von Terminals“ for encryption and keyexchange

- **PIN management**

Changing PINs

- **Geldkarte** (phased out in 2025)

Geldkarte is for cashless payment of goods and services using an electronic purse. Removed Geldkarte features from the XFS4IoT specification.

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL’s written permission

Guest speaker - Synaforce

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL's written permission

synaforce

synaforce

XFS4IoT SP-Dev

XFS4Iot PoC

Dominik Haas, Project Manager

2026, March 3rd

Who is synaforce?

- Founded in 1992 as Hartl EDV and on the market since 2023 through the integration of other companies under the new brand and company name synaforce
- The company operates multiple locations offering highly specialized 360-degree services in the high-end data center sector
- CSP & MSP for highly spezialized IT service providers, software providers and end customers with complex individual outsourcing requirements

- 6 locations and 8 offices throughout Germany
- 2 data centers in Hofkirchen & Mainz
- 4 colocation sites in Munich, Nuremberg and 2 in Berlin
- 150+ employees

- Highly certified & technology-agnostic
- Core industries: Critical Infrastructures (CI), financial and healthcare sector, trade, manufacturing, public institutions

synaforce

Our Customer
for POC

- large private Bank in Germany
- 2000+ ATM Network
- Withdrawl, Recycling and NonCash devices

synaforce

Current Production Situation

Starting point

- Full *Fat-Client* ATM Installation
- Costly hardware upgrades and complexity of components
- EPP Keyloading via OPT as per DK Specification

synaforce

Required Changes

Technical Design

PoC Setup

- VM in different Network than ATM
- Usage of Standard ATM Installer
- Installation of XFS4IoTWrapper on ATM
- Installation of Standard VM
- Connection ATM and VM via XFS4Iot

synaforce

XFS4IoT PoC Setup

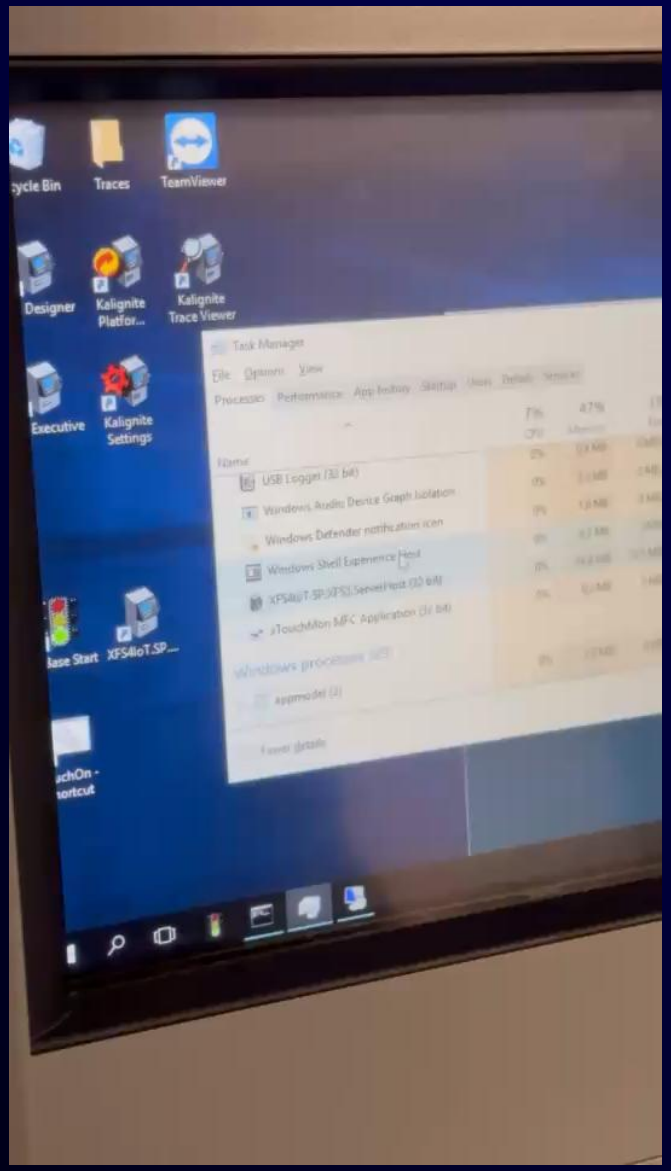
Technical Design

synaforce

XFS4IoT PoC

Sample Video

DNI Cineo2060



synaforce

XFS4IoT PoC

Sample Video
DNI Cineo2060

synaforce

XFS4IoT PoC

Trace Sample

incl. GermanSecureMessage

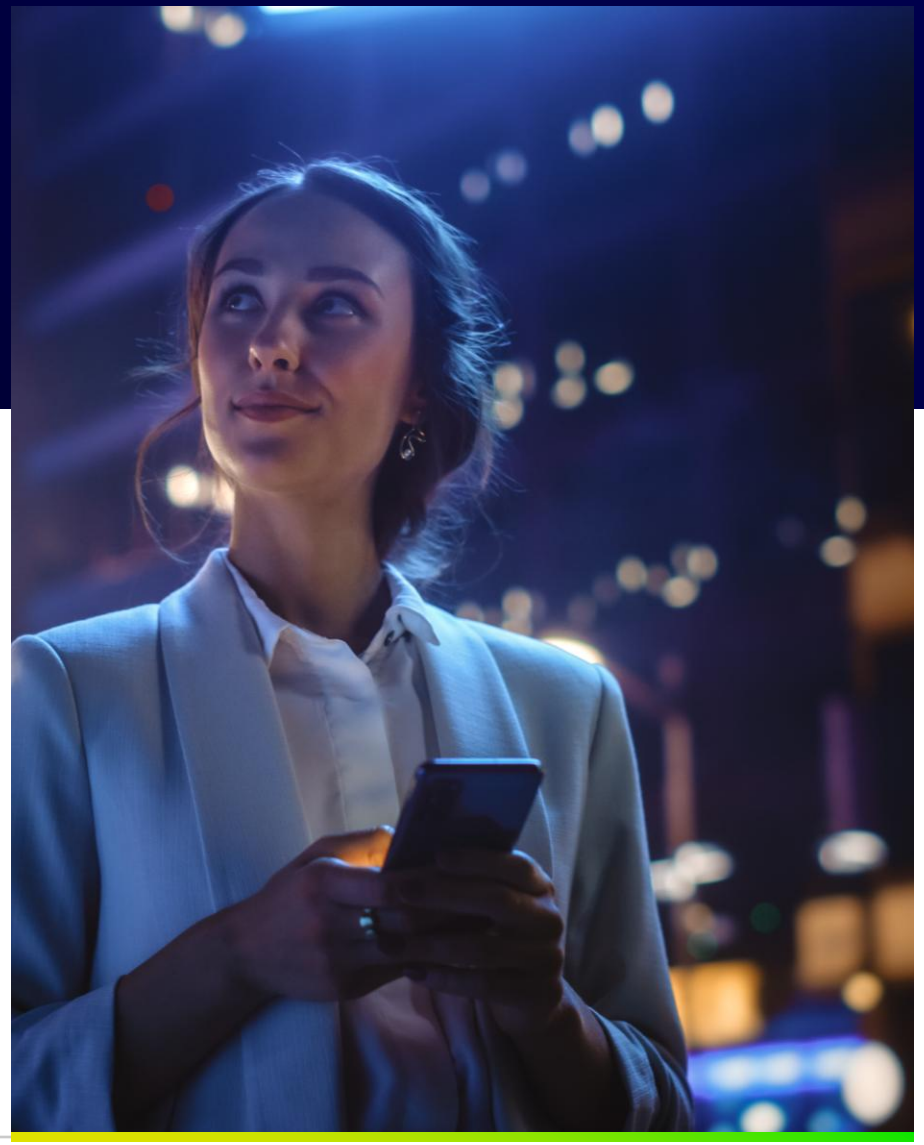
Trace Edit Find Bookmark Filter Options

ID	Component	SubSystem	Operation	Termin...	Transa...	Proce...	ThreadID	Event
0001C032	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: InService [Customer]
0001C3DC	CardRead	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00003DDC	ReceiveSync()->True: {"header":{"name":"Common.Status","requestId":67,"type":"completion","version":"3.0","completionCode":"success"},"payload":{"common":{"device":"online","devicePosition":"inPosition","powerSaveRe
0001C416	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: CardDetected [Customer]
0001C499	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: ChipDetected [Customer]
0001DFE9	CardRead	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00002E4C	ReceiveSync()->True: {"header":{"name":"CardReader.ChipIO","requestId":101,"type":"completion","version":"3.0","completionCode":"success"},"payload":{"chipProtocol":"chipT1","chipData":"6A-83"}}
0001E08B	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: PreparingChipApplicationAdvice [Customer]
0001E247	CardRead	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00002E4C	ReceiveSync()->True: {"header":{"name":"CardReader.ChipIO","requestId":110,"type":"completion","version":"3.0","completionCode":"success"},"payload":{"chipProtocol":"chipT1","chipData":"70-38-5F-24-03-26-12-31-5A-0A-
0001E46B	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: HostAuthorizationAdvice [Customer]
0001F343	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: HostAcceptedAdvice [Customer]
0001F597	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: SelectAmount [Customer]
0001FADB	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: HostAuthorizationAdvice [Customer]
0001FCA2	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: HostAcceptedAdvice [Customer]
0001FD75	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: ConfirmSurcharge [Customer]
0001FDAE	PinPad	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00003F1C	SendSync:{"header":{"name":"Keyboard.DataEntry","requestId":26,"type":"command","version":"2.0","timeout":0},"payload":{"maxLen":0,"autoEnd":true,"activeKeys":{"one":{"terminate":false},"enter":{"terminate":false},"canc
0001FE00	PinPad	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00003F1C	ReceiveSync()->True: {"header":{"name":"Keyboard.EnterDataEvent","requestId":26,"type":"event","version":"2.0"}}
0001FED3	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: EnterPin [Customer]
0001FED8	PinPad	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	000021EC	SendSync:{"header":{"name":"Keyboard.PinEntry","requestId":1124,"type":"command","version":"2.0","timeout":30000},"payload":{"minLen":4,"maxLen":12,"autoEnd":false,"echo":"*","activeKeys":{"zero":{"terminate":false},""
0001FF02	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: EnterPinUpdate [Customer]
0001FF21	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: EnterPinUpdate [Customer]
0001FF3B	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: EnterPinUpdate [Customer]
0001FF55	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: EnterPinUpdate [Customer]
0001FF62	PinPad	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	000021EC	ReceiveSync()->True: {"header":{"name":"Keyboard.KeyEvent","requestId":1124,"type":"event","version":"2.0"},"payload":{"completion":"enter","digit":"enter"}}
0001FF6E	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: EnterPinUpdate [Customer]
0001FFD9	PinPad	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	000021EC	ReceiveSync()->True: {"header":{"name":"Keyboard.PinEntry","requestId":1124,"type":"completion","version":"2.0","completionCode":"success"},"payload":{"digits":4,"completion":"enter"}}
00020068	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: ChipAuthorizationAdvice [Customer]
000200B0	CardRead	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00002E4C	ReceiveSync()->True: {"header":{"name":"CardReader.ChipIO","requestId":111,"type":"completion","version":"3.0","completionCode":"success"},"payload":{"chipProtocol":"chipT1","chipData":"77-37-9F-27-01-80-9F-36-02-08-
000201A9	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: HostAuthorizationAdvice [Customer]
0002022E	PinPad	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	000021EC	SendSync:{"header":{"name":"German.SecureMsgSend","requestId":1127,"type":"command","version":"1.0","timeout":300000},"payload":{"protocol":"genAs","msg":"02"}}
00020233	PinPad	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	000021EC	ReceiveSync()->True: {"header":{"name":"German.SecureMsgSend","requestId":1127,"type":"completion","version":"1.0","completionCode":"success"},"payload":{"msg":"81-06-22-B3-D6-62-28-DC-B6-F8-99-E6-1C-11-25-97-21-
00020278	PinPad	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	000021EC	SendSync:{"header":{"name":"German.SecureMsgSend","requestId":1128,"type":"command","version":"1.0","timeout":300000},"payload":{"protocol":"genAs","msg":"03-02-00-70-3C-46-C0-A0-E0-1E-89-F1-F0-67-22-42-00-01-10
0002027F	PinPad	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	000021EC	ReceiveSync()->True: {"header":{"name":"German.SecureMsgSend","requestId":1128,"type":"completion","version":"1.0","completionCode":"success"},"payload":{"msg":"08-34-B0-4D-F4-C0-53-6D"}}
0002058B	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: HostAcceptedAdvice [Customer]
000205CB	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: DispensingCashAdvice [Customer]
00020602	CashDisp	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00002330	SendSync:{"header":{"name":"CashDispenser.Dispense","requestId":63,"type":"command","version":"3.0","timeout":300000},"payload":{"denomination":{"denomination":{"app":{"currencies":{"EUR":20},"counts":{"53427SLOT-
000207DD	GuideLig	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00000B1C	ReceiveSync()->True: {"header":{"name":"Common.Status","requestId":68,"type":"completion","version":"3.0","completionCode":"success"},"payload":{"common":{"device":"online","devicePosition":"inPosition","powerSaveRe
000207E0	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: TakeCard [Customer]
0002094C	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: CardTakenAdvice [Customer]
0002095A	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: CardTakenAdvice [Customer]
00020968	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: PresentingCashAdvice [Customer]
00020BC7	CashDisp	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00002330	ReceiveSync()->True: {"header":{"name":"CashDispenser.Present","requestId":68,"type":"completion","version":"2.0","completionCode":"success"},"payload":{"position":{"position":"outDefault","additionalBunches":"0"}}}
00020BCD	CashDisp	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00002330	ReceiveSync()->True: {"header":{"name":"Common.Status","requestId":69,"type":"completion","version":"3.0","completionCode":"success"},"payload":{"common":{"device":"online","devicePosition":"inPosition","powerSaveRe
00020CA5	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: TakeCash [Customer]
00022333	CashDisp	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00002330	ReceiveSync()->True: {"header":{"name":"CashDispenser.GetPresentStatus","requestId":74,"type":"completion","version":"3.0","completionCode":"success"},"payload":{"denomination":{"currencies":{"EUR":20},"values":{"878
0002234B	CashDisp	OB_LAYER	XFS4COMM	00006B2B	00003B82	00003EE4	00002330	ReceiveSync()->True: {"header":{"name":"CashDispenser.GetPresentStatus","requestId":78,"type":"completion","version":"3.0","completionCode":"success"},"payload":{"denomination":{"currencies":{"EUR":20},"values":{"878
00022408	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: CashTakenAdvice [Customer]
00022C78	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: ThankYou [Customer]
00023CD6	K3A	USERINTF	EVENT	00006B2B	00003B82	00003EE4	00003F84	Customer UI event: InService [Customer]

What's next…

Customer Feedback

- Great starting point to migrate to fully native XFS4IoT setup
- Improves interoperability cost efficiently of different Hardware components
- Detaches dependency to Vendor XFS4IoT SPs
- Decreases time for application rollout cycles by virtualisation of application
- Further steps to come



Q & A

Dominik Haas, Project Manager

dominik.haas@synaforce.com

What's next?

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL's written permission

What’s next?

- Quarterly meetings moving forward

- Framework updates and roadmap
- More PoC and experimental work
- Guest speakers
- More demos (biometrics and more)

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL’s written permission

Next call

Zoom

Meeting once every quarter
at 1300 UK time for 30 mins

Next call: 2nd June 2026
1300 UK, 0800 US EST, 2100 Tokyo time

Calls are 30 mins long

We will continue to use Zoom
(Interpretation in Japanese, Chinese and Spanish is available using Zoom's interpretation feature)

© 2026 KAL ATM Software GmbH (KAL)

Confidential. This document must not be copied or distributed without KAL's written permission